
The effects of competition on the quality of primary schools in the Netherlands

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Abstract. We investigate the impact of competition between primary schools on the quality of education in the Netherlands. Do schools facing more competition in their neighbourhood perform better than schools facing less competition? As a measure of school quality, we look at the performance of pupils at the nationwide standard test (the so-called ‘Cito test’) in the final year of primary education. Since competition is likely to be endogenous to the quality of schools, we use the distance between the school and the city centre as an instrument for the level of competition faced by a school. Using a large range of data on pupil, school, and market characteristics, we find that school competition has a positive significant effect on pupil achievement, although the effect is very small. An increase in competition by one standard deviation (comparable to five additional schools in the market) increases pupil achievement as measured by the Cito test by 5% to 10% of the mean standard deviation, which is less than one point (on the Cito test scale from 500 to 550 points).

Keywords: education, competition, primary schools, pupil achievement

1 Introduction

In recent years, many countries have shown interest in public policies aiming to increase choice and competition in education. These policies are motivated by the free-choice argument, whereby parents should have good, safe, and affordable choices for educating their children, as well as by the standard economic argument that more competition would provide an incentive for schools to improve quality. To the extent that parents choose the best quality schools and that schools benefit from an increase in enrolment, schools will attempt to improve in order to retain and attract pupils.

The objective of the current paper is to investigate the impact of school competition on the quality of Dutch primary schools, as measured by pupils’ educational outcomes. Our research question is: do schools facing high levels of competition in their neighbourhood perform better than schools facing less competition? The Netherlands stands out as one of the few countries with very large freedom of school choice. Yet, little is known about the effect of school competition on the quality of education in the Netherlands.

There is an extensive literature on the effects of school competition on educational outcomes, mainly in the US and more recently in the UK (Belfield and Levin, 2002; Gibbons et al, 2008; Hoxby, 2000). Overall, the results from this literature are mixed. While some

studies find a positive link between competition and educational outcomes, estimates in many other studies lack statistical significance.

On methodological grounds, the literature is divided between two strands. One strand evaluates policies aiming to extend school choice and the impact of these reforms on educational outcomes. The effects are then compared with a control group or counterfactual not affected by the policy. Because these policy reforms are localised, it is often difficult to generalise these results to other educational markets. Holmes et al (2003) investigate how the introduction of school choice in North Carolina affected the performance of traditional public schools, measured by standard performance tests. The results imply an increase of approximately 1% in achievement when a traditional school faces competition from a charter school. Hsieh and Urquiola (2003) analyse the effect of a school reform which expanded school choice in Chile in 1981, and find no evidence that choice improved average educational outcomes.

Our paper is connected to the second strand in the literature, which studies the effects of implicit variation in the level of school competition in a cross-section of markets. The main challenge of these studies is to establish a causal link between pupil achievement and school competition. While increased school choice and competition may impact school quality, a causal relationship running in the opposite direction is equally likely. Indeed, as high-quality schools grow larger, they may become more and more monopolistic in the market. This problem of ‘inverse causality’, also coined ‘endogeneity’, between school competition and educational performance poses some methodological challenges. Looking at forty-one empirical studies in the US, Belfield and Levin (2002) conclude that, if any, the gains from competition tend to be very modest: a one standard deviation increase in competition increases test scores by approximately 10% of a standard deviation. Hoxby (2000) analyses the effect of school choice on schools’ productivity and sorting of students. Using an instrumental variables estimation approach, in which instruments are derived from the natural boundaries (rivers and streams), she concludes that “metropolitan areas with greater Tiebout choice have more productive public schools and less private schooling” (page 1209). Looking at data for England, Gibbons et al (2008) apply ordinary least squares (OLS) and instrumental variables (IV) estimation approaches, where they generate instruments for choice and competition in terms of distance from the educational market boundary.⁽¹⁾ IV estimates show no evidence of either more choice or more competition among schools improving pupils attainment.

This paper contributes to the existing literature by providing new evidence on the causal relationship between primary school competition and pupils’ achievement in the Netherlands, a country with a long tradition of free school choice. Dijkgraaf et al (2009) look at the relationship between competition and quality of education in Dutch secondary schools. These authors report only OLS estimation results and are thus unable to address the reverse-causality problem between school competition and school quality. Himmeler (2009) considers the effect of Catholic schools’ competition on non-Catholic schools’ performance and finds a positive effect of competition on academic achievement.

We define geographical markets for school competition and estimate the effect of competition on pupil performance measured by standard test scores at the end of primary school (the so-called ‘Cito test’). In order to correct for the endogeneity of our competition variable, we use the distance between the school and the city geographical centre as an instrument for school competition. The intuition is that schools located close to the city centre face more competition than schools located at the boundary of the city, because schools located close to the city centre are accessible to a larger number of pupils.

⁽¹⁾ Gibbons et al (2008) assume educational markets to be defined within the boundaries of a Local Education Authority (LEA), benefiting from the fact that “only a small percentage of pupils in England attend Primary schools outside of their home Local Education Authority because there are institutional barriers to doing so” (page 913).

Our empirical results suggest that pupils enrolled in schools facing more competition in their neighbourhood perform better than pupils in schools facing less competition. We estimate a small positive effect of competition on pupils' achievement. A one standard deviation increase in competition leads to an increase of 5–10% of the mean standard deviation of the Cito score (less than 1 point on a scale from 500 to 550 points). This result is robust to a large range of different specifications.

The paper is organised as follows. In section 2 we describe the education system in the Netherlands. In section 3 we present the main methodological issues and our empirical strategy and in section 4 we describe the data and sample construction. In section 5 we present our main results concerning the effects of competition on the Cito scores and some robustness checks. Section 6 concludes.

2 School choice and competition in Dutch primary schools

There are about 7000 primary schools in the Netherlands. Dutch education distinguishes two systems: the public schools (about 30% of all schools); and the publicly funded private schools. In private education a distinction is made between Protestant schools, Roman Catholic schools, and other private educational institutions. All religious groups and other groups representing certain philosophies of life—called 'denomination'—are free to start their own school and are, up to minimum standards, free to decide about didactics of the school. If there are a sufficient number of parents in a community who want to send their children to a public school, they can force the local government to open one.

In the Netherlands all public and private schools are equally financed by the government based on the number and distribution of pupils. Pupils are weighted on the basis of a number of criteria. Up to 2006, which is the period relevant for our analysis, the weighting scheme was as follows: (1) children from a Dutch cultural background whose parents have low level of education: a weighting of 0.25; (2) children of barge operators: 0.40; (3) children of caravan dwellers and gypsies: 0.70; (4) children from a non-Dutch cultural background whose parents have a low level of education and low-skilled occupations: 0.90; (5) all other children: no weighting. This weighting scheme implies that for each pupil weighted 0.25, 0.40, 0.70, or 0.90, a school gets respectively 25%, 40%, 70%, or 90% extra funding. In addition, schools receive extra personnel and other resources on the basis of these weightings.

Primary education is free since schools which are funded are not allowed to require extra payment from parents. Public schools are not allowed to refuse children who want to come to the school. Officially, private schools may require that parents, pupils, and teachers accept the principles on which the school is founded. This requirement is in most cases purely a procedural process. Indeed, many public schools nowadays offer religious classes, and a growing number of 'religious' schools are open to 'nonreligious' children. Additionally, the difference between teaching methods tends to become less visible. Parents have complete freedom of school choice, since all public and private schools are free of charge and there are no geographic or income restrictions.⁽²⁾

Several papers have looked at the motives of parents in their choice of school in primary education (Herweijer and Vogels, 2004; Karsten et al, 2002). Survey results indicate that parents state the 'quality of education' as the most important determinant of a school choice and look at a wide range of indicators to assess quality: results in the Cito test, the

⁽²⁾ In practice, in certain specific cases, the freedom of choice might be limited due to different factors. In the centre of some large cities, schools may be capacity constrained. Also, in certain neighborhoods in Amsterdam a postcode policy was recently adopted: only children living in the same postcode as the school can register at the school. However, such policies are relatively recent and they did not affect the sample of pupils analysed in this paper. In addition, in our robustness analysis, our results remain unaffected when we exclude the main large cities in the Netherlands.

reputation of the school, the number of pupils who continue into secondary education, the quality assessments of the Inspectorate of Education, etc. Remarkably, parents from a Dutch background also put weight on the level of segregation at the school, with higher perceived quality coinciding with fewer pupils from a non-Dutch background. In the surveys, this comes out under the label ‘matching between school and home’ (‘our type of people’), ‘atmosphere at school’, and ‘identification with the school’ (Karsten et al, 2002).

After quality, another important determinant of school choice is the distance to school. Most parents choose a school in their immediate neighbourhood. A survey of the travel behaviour from home to school of children in primary education found that the average distance that children travel is 1.26 km (Van der Houwen et al, 2004). Finally, parents’ choice is often irrespective of denomination. In 1990, 7% of the parents had no preference for the denomination of the school. In 2000 this share increased to 20%. In 1999, 40% of the parents sent their children to a school whose denomination did not match the parents’ religious background (Herweijer and Vogels, 2004).

Schools compete with other schools for pupils in a relevant market. Given the current financing scheme which is based on the number of pupils, schools have incentives to attract more pupils in order to receive more funding. In addition, with more pupils the chance that public financing may be stopped because the school is too small (the so-called ‘closing-down’ norm) decreases. The salary scale of school directors also depends on the size of the school. According to this financing scheme, pupils are thus valuable assets for a school and its management.

3 Empirical approach

3.1 Defining school competition

In this paper we define the concept of ‘school competition’ in a spatial context. We assume that a school competes with all other schools located within a radius of 1.5 km around the school. Given the way we construct our markets, in the case of connected cities the number of competing schools also includes schools which are located in adjacent cities.

Within each school competition market, we construct the ‘competitors’ variables as the number of alternative schools within a fixed radius of 1.5 km around a school. Monopoly markets thus have zero alternative competing schools within a school market. As a measure of competition within a school market, we also define the inverted Herfindahl index (HHI):

$$HHI_m = 1 - \sum_{i=1}^n s_{im}^2, \quad (1)$$

where s_{im} is the market share of a school i in a market m (ie, school pupil population related to total pupil population in the market), and n is the total number of schools in a market m . For ease of interpretation of the results, we invert the Herfindahl index such that a lower (higher) value indicates lower (higher) competition. A value of 0 indicates a monopoly market.

Our school market definition is in line with the evidence that primary school availability and accessibility in the Netherlands is very good. According to recent data from Bunschoten (2008) the nearest primary school is located, on average, 600 m from a pupil’s residence, which is less than 10 minutes walk; about 90% of children live less than 1 km and 60% less than 500 m away from a school. As a result, parents are not likely to make long journeys to bring their children to school because they have a large choice of schools in their neighbourhood.

On average, there are six schools within a fixed radius of 1.5 km around each school. Our definition of spatial competition in school markets is likely to be affected by other factors such as city size and urbanisation (ie, highly urbanised markets will have a higher density of schools than less urbanised markets). We control for these factors in our analysis.

In other approaches in the literature, school markets are defined as metropolitan areas (Hoxby, 2000; Hsieh and Urquiola, 2003), or more precise school catchment areas are defined when data on pupil residential location are available (Gibbons et al, 2008; Harris and Johnston, 2008). There is a large literature on the definition of spatial competition in geographical markets (using metropolitan areas, fixed radius distance, or journey times) related to measuring competition among hospitals (for reviews, see Propper et al, 2004; Wong et al, 2005).

Due to data constraints, we can only measure distances between the centres of four-digit postcode areas. This implies that we have to deal with potential measurement error in our empirical analysis. More precise distance data (for instance, between two schools or between two centres of six-digit postcode areas) were not available. As a consequence, our circles of 1.5 km are, in fact, circles around the centre of the four-digit postcode area in which the school is located. As an illustration, a school A is said to compete with a school B if the distance between the two postcode centres of the two schools A and B is within 1.5 km, even though school A might in reality be located more than 1.5 km away from school B. Inversely, a school C will not be considered as competing with school B if the distance between the two postcode centres of both schools B and C is larger than 1.5 km, even though school C might in fact be located less than 1.5 km away from school B. This implies that, for certain markets, competition will be underestimated (as in the case of schools B and C) and for other markets overestimated (as in the case of schools A and B). We come back to this issue below, when we discuss our empirical strategy.

3.2 Measuring school quality

When choosing a primary school, parents compare the quality of the different schools in their neighbourhood. About 80% of Dutch schools participate in the nationwide uniform Cito test in the final year of primary school. The Cito test measures the skills acquired by pupils in four areas: language, arithmetic, information processing, and world orientation (optional). The total score ranges from 500 to 550 points, with an average of 535 points and a standard deviation of 10 points. Although the test is not a formal graduation exam, pupils have incentives to perform well since their performance on the Cito test is one of the acceptance criteria for different levels of secondary education. Average Cito scores at the school level are, in principle, public information and are often available in folders and on websites published by the schools.

As a proxy for school quality, we look at the performance of pupils attending the school on the Cito test. There is evidence that Cito scores are correlated with many other aspects that matter to parents, such as the level of segregation or the percentage of pupils who continue into secondary education. Many studies show that parents are not indifferent to standardised test scores. Evidence from house prices shows that parents take average pupil achievement at schools into consideration when choosing their residential location (Black, 1999; Black and Machin, 2011; Kane et al, 2006). Using UK data, Gibbons and Silva (2011) show that parental perception of educational excellence is also related to standard test scores. Even though we cannot directly generalise these results to the Netherlands, evidence in the literature supports the use of test scores as a measurement of school quality. In addition, in order to measure the impact of school competition on the added value of a school, we adjust the Cito scores to account for the distribution of pupils within the school (ie, the share of pupils with a non-Dutch background). Ideally, the added value of a school is best measured by correcting for prior educational achievement (when the pupil enters the school). In our case, however, pupils have not been pretested. Therefore, we only consider 'contextual added value' of schools, in which test scores are only corrected for pupils' socioeconomic background (Thomas and Mortimore, 1996).

We estimate the effect of competition on the quality of schools, measured by pupils' performance on the Cito test, correcting for pupil, school, city, and neighbourhood characteristics. The following specification defines our empirical approach:

$$\text{school performance}_{ismt} = \alpha_0 \text{competition}_{mt} + \mathbf{x}_{ismt}^T \boldsymbol{\alpha}_1 + \varepsilon_{ismt}, \quad (2)$$

where $\text{school performance}_{ismt}$ is the Cito test score for pupil i , attending school s , located in market m , at time t ; competition_{mt} is a competition index (the number of alternative schools, or the Herfindahl index) in market m , at time t ; $\mathbf{x}_{ismt}^T$ is a vector of pupil, school, city, and neighbourhood characteristics; ε_{ismt} is an error term clustered at the postcode level.

3.3 Instrumental variables (IV) strategy

One of the main issues when measuring the effects of competition on school quality is that the level of competition observed in a market may be endogenous to the quality of schools. Thus, a correlation between competition and school performance does not necessarily imply a causal link. This problem occurs because families tend to consider the quality of schools when deciding in which neighbourhood to live. As a result, the level of competition that we observe in the market might be endogenously determined by the quality of schools in the market. Residential sorting might, for instance, lead families to crowd around high-quality schools, such that high-quality schools will tend to be larger than low-quality schools. As high-quality schools grow larger, this reduces the apparent level of competition in the market. We therefore face a problem of endogeneity in the form of *reverse causality* between competition and the quality of schools. In addition, due to the selection of parents into certain neighbourhoods, there might be unobserved factors affecting the pupils' performance which may also be correlated with competition, causing competition to be endogenous to the quality of schools. This could occur if socioeconomic characteristics or preferences of families, which are correlated with school quality, are also related to the level of school competition in the market. For instance, if parents of pupils performing well at school are dissatisfied with the quality of available schools, they can start up an alternative school, thereby increasing the level of competition in the market.

The implication of endogenous competition is that the ordinary least squares (OLS) estimate of coefficient α_0 next to the competition variable in the regression in equation (2) will be biased. Hence, instead, we adopt an IV strategy in which we use an instrument as a proxy for our competition variable. This instrument should be correlated with the observed level of competition in the market, but not with the quality of schools. We use the distance between the school and the city geographical centre as an instrument for competition: the closer (further away) schools are to the city centre, the more (less) competition they face.⁽³⁾ The intuition is that families are most likely to go to the nearest schools: parents living at the city boundary will enrol their children in schools located further away from the city centre, whereas parents living close to the centre of the city will enrol their children in schools in that neighbourhood. Families living at the periphery of the city face longer journeys, and thus higher travel costs, to go to any school other than the nearest. As a consequence, schools located close to the city boundary (which are further away from the highly populated city centres) face less competition than do schools located at the centre of the city (which are easily accessible to a large number of parents). This approach relates to the study of Gibbons et al (2008), who use the distance to the market boundary as an instrument for competition in the school markets in England.

⁽³⁾We use the geographical centre, which is calculated based on the city administrative boundaries. This does not necessarily correspond to the historical centre or to the business activity centre of the city.

This IV strategy rests on several assumptions. Firstly, school competition should decrease with increasing distance from the city centre. In other words, there should be fewer schools at the periphery of the city than in the centre. In addition, after correcting for observable characteristics, the distance between the school and the city centre should not be correlated with the quality of schools (or other unobserved variables affecting the quality of schools). These are empirical issues which are addressed when we test the validity of our instrument in section 5.

There is an additional data limit due to the lack of precision of our distance measures. As noted above, we only measure distances between the centres of two four-digit postcode areas. In our dataset, all schools located in the same postcode area will therefore be the same distance from the city centre wherever their actual location is. In practice, this does not need to affect our estimates as long as we have a good instrument (Card, 1999, page 1821).

4 Data and sample construction

We obtained Cito scores of pupils who took the Cito test over the period 1999 to 2003 directly from the Cito organisation. The dataset includes the standard total Cito test score as well as the component scores on language, arithmetic, and information processing. The total Cito score and its components are standardised, with mean 0 and standard deviation 1. We excluded the bottom and top 1% of the pupils to avoid potential outliers. The only additional information that we have at the pupil level is gender.

Next, we obtained several datasets from the Inspectorate of Education, including data on the addresses of all primary schools, the composition and size of the pupil population in each school, and the number of teachers.⁽⁴⁾ We also obtained a dataset from the Central Funding of Institutions (CFI), which describes the composition of school boards over the same period. Regarding school denomination, we made a distinction between four groups: public schools, Catholic schools, Protestant schools, and other private schools. We dropped some specific private schools—namely orthodox Protestant schools (reformed and evangelistic), Islamite, Hindu, and Jewish schools—as we cannot reasonably assume that these schools compete with the other public and private schools.

In our dataset about 15% of schools have more than one building location. For these schools, we assumed that school performance is uniform across school locations and that, when a school has several locations within the same market, these school buildings are not competing with one another.

In total, we constructed about 6000 school competition markets (defined as a geographic market within a radius of 1.5 km around every school) distributed over 3000 four-digit postcode areas in about 1600 cities. We obtained demographic data at the four-digit postcode area and city level from Statistics Netherlands. To correct for market characteristics, we included demographic variables of the four-digit postcode in which the school is located. We also control for the size and urbanisation at the city level; this is mainly to correct for a scale effect in our instrumental variable (the larger the city, the larger the distance to the city centre). In addition, we used data on distance variables between two centres of four-digit postcodes and between the centre of a four-digit postcode and the city geographical centre.

Our sample was restricted to cities with more than 20 000 inhabitants. For comparison, Hoxby (2000) analysed metropolitan areas and Gibbons et al (2008) looked at greater London. In small rural cities with low school densities our competition indices are very highly correlated with urbanisation and city-size variables, leading to multicollinearity problems. In larger markets we find more mixed patterns of competition, city size, and urbanisation levels.

⁽⁴⁾ Because the data on teachers are missing for the years 1999 and 2003, we conducted the analysis on the 2000–02 and on the 1999–2003 (excluding data on teachers) samples. This did not affect our estimates of the effects of competition.

Table 1. Descriptive statistics, cities with $\geq 22\,000$ inhabitants, 1999–2003.

	Mean	Standard deviation (SD)	Minimum	Maximum	Observations
Pupil characteristics					
Cito (mean = 0, SD = 1)	0	1	−2.5	1.6	358 767
Gender (0 = female, 1 = male)	0.5	0.5	0	1	358 767
School characteristics					
Number of pupils (in 100s)	3.24	1.43	0.20	12.38	358 767
% pupils weight 0.90	0.2	0.26	0	1	358 767
Size of school board ^a	5.67	2.94	1	10	358 767
Public schools (dummy)	0.33	0.47	0	1	358 767
Catholic schools (dummy)	0.33	0.47	0	1	358 767
Protestant schools (dummy)	0.25	0.43	0	1	358 767
Other schools (dummy)	0.09	0.28	0	1	358 767
Market characteristics					
<i>Postcode level variables</i>					
% population with high income	0.36	0.06	0.09	0.59	358 767
% population with low income	0.45	0.06	0.29	0.72	358 767
% population under 14 years old	0.18	0.04	0.02	0.31	358 767
<i>City level variables</i>					
City size (in 100 000s inhabitants)	1.57	1.91	0.21	6.56	358 767
City urbanisation ^b	3.2	0.7	2	4	358 767
Competition variables (1.5 km)					
Number of alternative schools	6.5	4.8	0	28	358 767
Herfindahl index (inverted)	0.78	0.16	0	0.96	358 767
Distance to city centre	2.6	2.1	0	23	358 241
Number of alternative public schools	2.4	2.7	0	13	118 208
Number of alternative Catholic schools	1.8	1.6	0	8	101 274
Number of alternative Protestant schools	2.0	1.9	0	12	88 764
Number of alternative schools belonging to a different school board	3.1	2.1	0	14	117 491
Competition variables (2.5 km)					
Number of alternative schools	14.7	10.1	0	62	362 090
Herfindahl index (inverted)	0.89	0.1	0	0.98	362 090

^aDeciles of number of schools in school board: 1 = very small; 10 = very large.

^bQuartiles of number of addresses per km²: 1 = very low urbanisation; 4 = very high urbanisation.

We were left with a sample covering about 50% of all primary schools in the Netherlands and 60% of all pupils. Figure 1 shows the distribution of the number of alternative schools within our geographical markets.

Table 1 provides the descriptive statistics of our sample dataset. We have a repeated cross-sections dataset of about 360 000 individual pupils who passed the Cito test over the 1999 to 2003 period. On average, there are about 6.5 alternative schools within a school market (the standard deviation is 5 schools). The inverted Herfindahl index is 0.78 on average, reflecting a high level of competition.

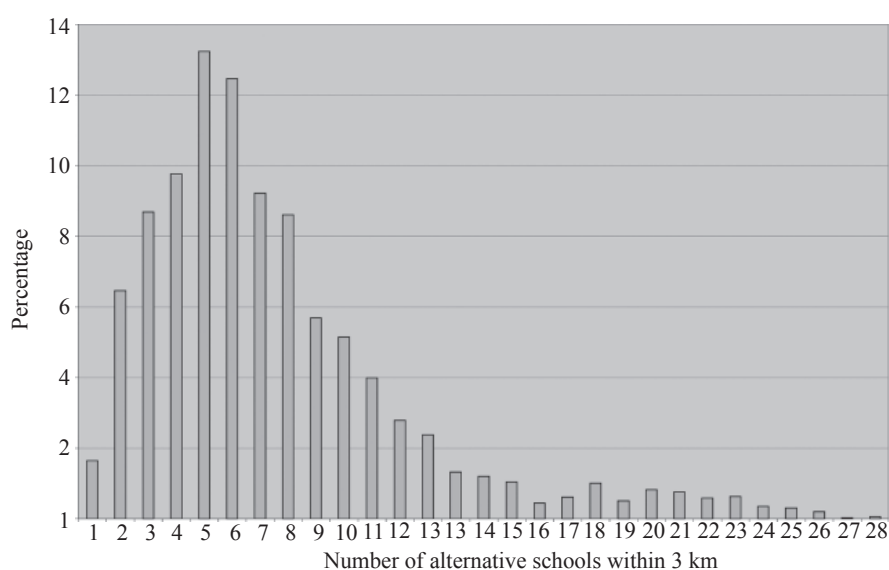


Figure 1. Distribution of the number of alternative schools within a 1.5 km radius, ie, 3 km diameter (percentage of markets with 0,1,..., 28 schools), cities with $\geq 20\,000$ inhabitants.

5 Results

5.1 Basic specification

Table 2 summarises the effects of competition on pupils' achievement, which is the coefficient of interest in this study. Table 2(a) gives the results of IV estimation of the effect of competition on the total Cito score of each pupil, as well as on each component of the Cito score: language, arithmetic, and information processing. In our IV estimation approach we instrument the competition variable with the logarithm of the distance between the school and the city centre. The results are presented for both competition variables: the number of competing schools and the (inverted) Herfindahl index. Columns (1) and (2) in table 2(a) give the baseline specification used in the remainder of this study.

We find evidence of a small positive effect of school competition on educational outcomes, significant at the 10% level. Pupils in schools facing more competition in their neighbourhood perform, on average, better than pupils in schools facing less competition. In columns (1) and (2) in table 2(a) an increase in competition by one standard deviation leads to an increase in the Cito score by 4.7% and 7.5% of a standard deviation when we measure competition using the number of alternative schools and the Herfindahl index, respectively. When we include city fixed effects, the effects are slightly lower, as shown in columns (3) and (4) in table 2(a).

The general result over all specifications in table 2(a) is that the gains from competition tend to be in the range from 5% to less than 10% of a standard deviation in pupil achievement. Put differently, an increase of one standard deviation in competition leads to an increase of 5–10% of a standard deviation in the Cito score, which is less than 1 point on the scale from 500 to 550 points (the standard deviation of the Cito score on all pupils in the Netherlands is 10 points). This effect seems modest since increasing the level of competition in the market by one standard deviation is comparable to having about five additional schools in the market. Therefore, a large increase in competition is needed to raise the level of pupil achievement by one extra point.

Table 2(b) gives the results of OLS estimations. In these specifications the effects of endogenous competition are not corrected for. In this case we find no significant effect of competition. In addition, whereas the IV results show that the exogenous variation in competition has a small positive effect on school achievement, our OLS estimates are much smaller than our IV estimates. Indeed, the remaining endogenous variation in competition is

Table 2. Baseline estimates with robust standard errors clustered at the postcode level shown in parentheses, instrumental variables (IV) and ordinary least squares (OLS) results; dependent variable = Cito score per pupil; cities with $\geq 20\,000$ inhabitants.

	Cito total									
	(1)	(2)	(3)	(4)	Cito language (5)	Cito arithmetic (6)	Cito information (7)	Cito language (8)	Cito arithmetic (9)	Cito information (10)
(a) IV estimation results										
Number of alternative schools	0.047* (0.027)		0.034* (0.018)		0.054** (0.026)	0.027 (0.028)	0.049** (0.027)			
Herfindahl index (inverted)		0.075* (0.043)		0.053* (0.030)				0.086** (0.042)	0.043 (0.045)	0.078** (0.037)
Controls	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
City fixed effects	no	no	yes	yes	no	no	no	no	no	no
Observations	358 137	358 137	358 137	358 137	358 137	358 137	358 137	358 137	358 137	358 137
(b) OLS estimation results										
Number of alternative schools	0.004 (0.007)		0.017 (0.007)		0.009 (0.007)	−0.001 (0.007)	0.003 (0.006)			
Herfindahl index (inverted)		−0.000 (0.009)		0.009 (0.007)				0.009 (0.009)	−0.012 (0.009)	0.003 (0.007)
Controls	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
City fixed effects	no	no	yes	yes	no	no	no	no	no	no
Observations	358 663	358 663	358 663	358 663	358 663	358 663	358 663	358 663	358 663	358 663

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Note: Competition variables are standardised with mean 0 and standard deviation 1.

negatively linked with test scores due to endogenous pupil sorting across schools. When the most able pupils are concentrated in a few schools, these schools become larger (and better) and the level of competition in the market becomes lower.

Looking at the different components of the Cito score [columns (5) to (10) in table 2(a)], the largest effect of competition is found for the language part of the Cito test.

Table 3 shows the effects of other factors in baseline specifications explaining pupil achievement, such as pupil, school, and neighbourhood characteristics. All estimated coefficients have the expected signs. On average, large schools perform better than

Table 3: Baseline specification with robust standard errors clustered at the postcode level shown in parentheses, instrumental variables (IV) results; dependent variable = total Cito score per pupil; cities with $\geq 20\,000$ inhabitants.

	Cito total	
	(1)	(2)
<i>Competition</i>		
Number of alternative schools	0.047* (0.027)	
Herfindahl index (inverted)		0.075* (0.043)
<i>Pupils</i>		
Gender (0=female, 1= male)	0.039*** (0.003)	0.039*** (0.003)
<i>School characteristics</i>		
Number of pupils	0.027*** (0.004)	0.028*** (0.004)
Percentage of pupils weight 0.90	-0.957*** (0.043)	-0.934*** (0.037)
Size of school board	-0.009*** (0.002)	-0.009*** (0.002)
Catholic schools	0.055*** (0.013)	0.058*** (0.013)
Protestant schools	-0.009 (0.013)	-0.009 (0.013)
Other schools	0.129*** (0.021)	0.130*** (0.021)
<i>Market characteristics</i>		
City size	0.001 (0.007)	0.006 (0.005)
City urbanisation	-0.058*** (0.011)	-0.061*** (0.012)
Percentage of population with high income	1.082*** (0.114)	1.052*** (0.111)
Percentage of population with low income	-0.353** (0.153)	-0.330** (0.152)
Percentage of population under 14 years	-0.788*** (0.088)	-0.753*** (0.170)
Observations	358 137	358 137

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Note: competition variables are standardised with mean 0 and standard deviation 1.

small schools. This could be explained by scale effects, as large schools might benefit from better teaching practices. Schools with a large share of pupils with a non-Dutch background perform, on average, worse than schools with a small share of minority students. Schools falling under larger school boards seem to perform worse than schools falling under smaller school boards. In addition, Catholic schools perform better than public and Protestant schools. Pupils in schools located in wealthy neighbourhoods have higher Cito scores than pupils going to schools in less wealthy neighbourhoods. All these coefficients, however, should be interpreted with caution since they may not necessarily reflect causal links.⁽⁵⁾

5.2 Assessment of IV strategy

We instrument our competition variables with the logarithm of the distance between school and the city geographical centre. In order to assess our IV strategy, we look at the results of the first-stage estimates of the IV regressions, where we estimate the effect of the logarithm of the distance between school and city centre on the level of competition in our school markets. We expect a significant negative effect if competition does indeed decrease with the distance from the city centre.

Column (1) in table 4 shows the results of the first-stage regression of the IV estimation procedure without controls; column (2) includes the same control variables as earlier (see table 3). We find that our instrument (distance school–city centre) is always very powerful in explaining the level of competition in the market. Areas with less school competition are located, on average, further away from the city centre than are areas with more school competition. A 10% increase in the distance between the school and the city centre reduces the number of alternative competing schools by 4.03% of a standard deviation. The *F*-test for the quality of the instruments is significant and always very high [>10 , which is the reference threshold for strong instruments (Staiger and Stock, 1997)]. The strong link between competition and our distance measure also suggests that potential measurement errors in our distance measures is not likely to affect our results. Next to distance to the city centre, other market and school characteristics also affect the level of competition in a market, as shown in table 4. School competition tends, for instance, to be higher in larger urbanised markets with a high share of young people.

The results in table 4 suggest that, on average, school competition decreases with the distance from the city centre. An important assumption in our IV strategy is that the distance to the city centre is not correlated with other unobserved characteristics of pupils, schools, and neighbourhoods which we are not controlling for but which may still affect pupil achievement. For instance, pupils living close to the city centre may be more motivated and may value test results more than pupils living at the boundary of the city. If the most able pupils live close to the city centre, then pupils' ability will be negatively correlated with the distance between the school and the city centre. Similarly, parents living close to the city centre differ from parents living at the city boundary with respect to education and income levels, but also in their educational investment in their children, and this in turn might affect pupils' performance.

We attempted to minimise the concerns about potential correlation between school quality and the distance measure by controlling for a large set of covariates in our baseline estimations. However, there might be other factors affecting pupils' performance which we do not observe. For this purpose we obtained additional data from Statistics Netherlands on

⁽⁵⁾Indeed, some of these variables, notably the number of pupils and the percentage of pupils with weight 0.90, might also be endogeneously determined by test scores. For instance, better schools tend to attract more pupils and will increase in size. Ideally, these variables should also be instrumentalised to investigate the true causal link with pupils' performance. Here, however, we only focus on the competition coefficient.

Table 4. First-stage estimates, with robust standard errors clustered at the postcode level shown in parenthesis; dependent variables = number of alternative schools/inverted Herfindahl index (HHI); cities with $\geq 20\,000$ inhabitants.

	Number of alternative schools		HHI	
	(1)	(2)	(1)	(2)
<i>Competition</i>				
Ln(distance school-city centre)	-0.423*** (0.033)	-0.403*** (0.033)	-0.271*** (0.022)	-0.253*** (0.022)
<i>Pupils</i>				
Gender (0 =female, 1 =male)		0.001 (0.003)		-0.001 (0.002)
<i>School characteristics</i>				
Number of pupils		-0.050*** (0.013)		-0.047** (0.009)
percentage of pupils weight 0.90		0.695*** (0.120)		0.128 ** (0.059)
Size of school board		0.020*** (0.005)		0.005 (0.004)
Catholic schools		-0.093** (0.045)		-0.108*** (0.029)
Protestant schools		0.095** (0.049)		0.042 (0.027)
Other schools		-0.045 (0.068)		-0.040 (0.040)
<i>Market characteristics</i>				
City size	0.332*** (0.017)	0.273*** (0.019)	0.098*** (0.006)	0.085*** (0.007)
City urbanisation	0.241*** (0.023)	0.238*** (0.025)	0.188*** (0.020)	0.196*** (0.021)
percentage of population with high income		-1.103*** (0.373)		-0.303 (0.181)
percentage of population with low income		0.161 (0.523)		-0.198 (0.286)
percentage of population under 14 years		1.240** (0.665)		0.318 (0.336)
Observations	371 790	358 137	371 790	358 137
R^2	0.38	0.42	0.19	0.21
F -test on excluded instruments	743	725	680	618
Partial R^2 on excluded instruments	0.06	0.06	0.06	0.05

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Note: competition variables are standardised with mean 0 and standard deviation 1.

neighbourhood characteristics at the postcode level, such as average house prices, average income, and the percentage of people receiving social benefits. Table 5 reports the coefficients after regressing each neighbourhood attribute on the distance to the city centre. All data are aggregated at the four-digit postcode level. Column (1) reports the results without controls, while column (2) includes the same control variables as earlier (see table 3). We do not find

Table 5: Distance school–city centre and additional neighbourhood characteristics; cities with $\geq 20\,000$ inhabitants; robust standard errors, clustered at the city level, are shown in parentheses.

	(1)		(2)	
	no controls	observations	with controls	Observations
Ln (average house prices)	−0.02 (0.13)	1144	0.14 (0.10)	1062
Ln (average fiscal income per year)	−0.01 (0.07)	968	−0.001 (0.06)	904
Percentage of population receiving social benefits	0.14 (0.28)	1095	−0.25 (0.22)	1025

Note: Regression (OLS) coefficients obtained from separate regressions of each variable on log of distance school–city centre. Control variables are the same as in table 3, averaged at the four-digit postcode level.

a significant relationship between these additional neighbourhood attributes and our distance instrument.⁽⁶⁾

Finally, using additional data from the PRIMA survey, a project aiming to assess the development of primary education, we can provide further evidence that our instrument is not strongly correlated with other pupil characteristics. In particular, we show that pupils' ability and family-background characteristics are not correlated with the distance variable. After all, ability at the pupil and school levels are the main drivers of test scores. If all of the most able pupils attend schools that face more competition, our results will be biased.

We used the nationally representative sample of the PRIMA survey and obtained data on the IQ score and parents' education of pupils attending 600 primary schools over the 1994–99 period.⁽⁷⁾ We matched the average IQ of pupils attending a school in a given postcode with our demographic dataset, including distance to the city centre. Since the PRIMA survey uses different identification numbers for schools from those of our baseline Cito dataset, we could match both datasets only at the four-digit postcode level. Further, we also computed the average of all school characteristics per four-digit postcode area. This allows us to check the correlation between our instrument variable and average school ability at the postcode level.

After dropping missing values, we were left with about 400 postcode areas. Table 6 shows the results after regressing pupils' characteristics, such as pupils' IQ and parents' education on our distance measure. In column (1) city size and urbanisation variables are only controlled for in order to correct for scale effects in our distance measures (larger distances in larger cities); in column (2) all the covariates are added as in table 3. The coefficients on our distance measures are never significant, suggesting that IQ and parents' education are not correlated with the distance between school and city centre.

Next to data on pupils' characteristics, in our baseline Cito dataset, we also have data on the yearly outflow of teachers per school. Distance to the city centre could be correlated with the outflow of teachers if, for instance, schools at the periphery had different working

⁽⁶⁾Based on anecdotal evidence, we would have expected to find that the wealthier people in the Netherlands tend to live in the city centre. Instead, the data reveal that there is no significant relationship between distance to the city centre and average wealth in the neighbourhood. A potential explanation is that we consider the geographical centre, rather than the 'historical' centre of the city.

⁽⁷⁾Although these data cover an earlier period than the period studied in our baseline analysis, geographical sorting of pupils is not likely to have changed much over the subsequent period. IQ of pupils is computed on a scale up to 30 points. The education level of parents is defined as follows: 1—low education, 2—low vocational education, 3—middle vocational education, 4—high (vocational) education.

Table 6. Estimates of effects of $\ln(\text{distance school-city centre})$ on pupils' characteristics; robust standard errors, clustered at the postcode level, are shown in parentheses.

	IQ pupil		Education father		Education mother	
	(1)	(2)	(1)	(2)	(1)	(2)
$\ln(\text{distance school-city centre})$	0.051 (0.158)	-0.057 (0.144)	0.060 (0.058)	0.071 (0.048)	0.030 (0.056)	0.026 (0.047)
City size and urbanisation variables	Yes	Yes	Yes	Yes	Yes	Yes
All control variables	No	Yes	No	Yes	No	Yes
Observations	472	450	462	441	461	440
R^2	0.17	0.29	0.18	0.46	0.19	0.44

Notes: Regression (OLS) coefficients obtained from separate regressions of pupil characteristics on log of distance school-city centre. Observations at the four-digit postcode level area. Controls in columns (1) include city size and urbanisation variables. Controls in column (2) include the same controls as in table 3, averaged at the four-digit postcode level (except for city size and urbanisation variables).

conditions (or working 'atmosphere') and management practices from those in schools in the city centre. Here again, we did not find evidence that the yearly outflow of teachers is correlated with our instrument.

Although we cannot completely rule out the possibility that our distance measure may be correlated with other unobservables which we cannot measure, the findings of our additional tests lend some support for the validity of our distance school-city centre instrument, as well as to the instrumental variables estimation results presented in this paper.

5.3 Robustness checks

To check the robustness of our results, we ran additional specifications. We tested whether there are nonlinear effects of our measures of competition on pupils' performance, but found no significant results.⁽⁸⁾ We conducted estimations on samples differentiated by the city size, and found that the effect of competition on the Cito test is more important in larger cities.⁽⁹⁾ This could be explained by the fact that the intensity of competition might be higher in large cities. The results are also not affected by excluding all markets in the four largest cities in the Netherlands (Amsterdam, Rotterdam, The Hague, and Utrecht).

Table 7 show the results obtained when we tested different assumptions on the way in which schools compete with one another. In column (1) we assume that public schools compete only with other public schools. In columns (2) and (3) we assume that Catholic and Protestant schools only compete with other Catholic and Protestant schools, respectively. The only significant positive effect of competition on pupil outcomes was found in the case of Catholic schools. In the literature, higher incentives for Catholic schools are often attributed to better governance and religious fervour. In column (4) the assumption is that schools compete only with schools falling under a different school board. Using the school-board definition of school competition, estimates of the effects of competition are larger than in

⁽⁸⁾ Inclusion of a squared term for our competition measure gave no significant results. Conducting estimations on two subsamples of small markets (below the median of 6 schools per market) and large markets (above 6 schools) yielded similar results (results available on request). This might be surprising since we could expect that the entry of a new school in a monopoly market would have greater effects than the entry of a new school into a market with ten schools. Yet denomination and differentiated teaching methods might play a greater role in smaller markets than in larger ones, thereby reducing the effects of competition in these markets.

⁽⁹⁾ Results available on request.

Table 7. IV estimation results—different definitions of school competition; dependent variable = total Cito score per pupil.

	Public (1)	Catholic (2)	Protestant (3)	School boards (4)	2.5 km (5)
Number of alternative schools	0.043 (0.064)	0.057** (0.030)	−0.052 (0.047)	0.054* (0.029)	0.095** (0.045)
Herfindahl Index (inverted)	0.049 (0.074)	0.079** (0.049)	−0.057 (0.049)	0.092* (0.050)	0.131** (0.061)
Observations	117 681	117 697	88 764	361 564	358 241

Notes: The same variables are controlled for as in the baseline specification in table 3.

First stage results always show a strong negative effect of distance school–city centre on competition measure.

In column (4) we count all public schools as one school board (as all public schools are administrated by the local authority); in column (5) we measure the number of alternative schools within a fixed radius of 2.5 km around each school.

our baseline specification—suggesting that schools might indeed have more incentives to compete with schools from a different school board. In column (5) the assumption is that schools compete with all other schools located within a 2.5 km radius around each school. In this case, school markets are much larger with, on average, 14 competitors per market. Estimated effects of school competition on school performance in 2.5 km defined markets are an upper bound of the found effects.

6 Conclusions

In this paper, we have examined the causal effect of school competition on pupils' achievement in Dutch primary schools. Our research question is: do schools facing more competition in their neighbourhood perform better than schools facing less competition? We estimated the effect of school competition on the Cito score at the pupil level, using an instrumental variable (IV) approach. We used the intuition that schools located at the periphery of a city are likely to face less competition than schools located close to the city centre, and instrumented school competition by distance between the school and the city centre. Assessment of the performance of our IV strategy revealed that this distance measure is a good instrument for the observed level of competition in the market.

We found a positive significant effect of competition on pupils' achievement in our IV framework. However, the effects are very small. A one standard deviation increase in competition increased the average Cito score by 5%–10% of a standard deviation, which is less than 1 point on the Cito scale from 500 to 550 points. This result was robust to a large range of specifications. Gains from school competition on pupil achievement are thus very modest—a result commonly found in the international literature.

Why do the effects of competition on pupils' achievement tend to be so small? One possible explanation is that parents lack information about the quality of schools, thereby reducing the incentives for schools to compete. In this case, policies aiming to increase transparency of school performance might have beneficial effects on the quality of education. Although newspapers publish school rankings for secondary schools in the Netherlands, such tables are not yet available for primary education.

Another explanation of the small effect of competition on educational outcomes is that parents might value other factors, such as school activities and maintenance of buildings above the Cito scores when choosing a school. In this case, schools might be competing on aspects other than pupil achievement. Finally, since the Netherlands has a long tradition of

high levels of school competition, gains from further additional competition may also be limited. Future research should provide a better understanding of these mechanisms.

Nevertheless, the very modest gains from competition on pupil achievement are not negligible. It is well established in the literature that even small increases in pupil achievement can have important impacts in the long term (Jamison et al, 2007). Several studies have shown for instance that even a small increase in pupil achievement may have a large impact on the future earnings of pupils (Mulligan, 1999; Murnane et al, 2000). Hanushek and Kimko (2000) have also established a causal effect between higher pupil test scores and economic growth.

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References

- Belfield C R, Levin H M, 2002, "The effects of competition between schools on educational outcomes: a review for the United States" *Review of Educational Research* **72** 279–341
- Black S E, 1999, "Do better schools matter? Parental valuation of elementary education" *The Quarterly Journal of Economics* **114** 577–599
- Black S E, Machin S J, 2011, "Housing valuations of school performance", in *Handbook of the Economics of Education, Volume 3* Eds E A Hanushek, S J Machin, L Woessmann (North-Holland, Amsterdam) pp 485–519
- Bunschoten B, 2008, "Hoe ver woon ik van..." *Bevolkingstrends, 2e kwartaal 2008* Centraal Bureau voor de Statistiek, The Hague
- Card D, 1999, "The causal effect of education on earnings", in *The Handbook of Labour Economics, Volume 3A* Eds O Ashenfelter, D Card (North-Holland, Amsterdam) chapter 30
- Dijkgraaf E, Gradus R H J M, de Jong M, 2009, "Competition and educational quality: evidence from the Netherlands", DP TI 2009-100/3, Tinbergen Institute
- Gibbons S, Silva O, 2011, "School quality, child wellbeing and parents' satisfaction" *Economics of Education Review* **30** 312–331
- Gibbons S, Machin S J, Silva O, 2008 "Competition, choice, and primary school performance" *Journal of the European Economic Association* **26** 912–947
- Hanushek E A, Kimko D D, 2000, "Schooling, labor-force quality, and the growth of nations" *American Economic Review* **90** 1184–1208
- Harris R, Johnston R, 2008, "Primary schools, markets and choice: studying polarization and the core catchment areas of schools" *Applied Spatial Analysis and Policy* **1** 54–80
- Herweijer L, Vogels R, 2004 "Ouders over opvoeding en onderwijs", Social and Cultural Planning Office of the Netherlands, The Hague
- Himmeler O, 2009, "The effects of school competition on academic achievement and grading standards", WP 2676, CESifo, Munich
- Holmes G M, DeSimone J, Rupp N G, 2003, "Does school choice increase school quality, WP 9683, National Bureau of Economic Research, Cambridge, MA
- Hoxby C M, 2000, "Does competition among public schools benefit students and taxpayers?" *The American Economic Review* **90** 1209–1238
- Hsieh C T, Urquiola M, 2003, "When schools compete, how do they compete? An assessment of Chile's nationwide school voucher programme", WP 10008, National Bureau of Economic Research, Cambridge, MA
- Jamison E A, Jamison D T, Hanushek E A, 2007, "The effects of education quality on income growth and mortality decline" *Economics of Education Review* **26** 772–789
- Kane T J, Riegg S K, Staiger D O, 2006, "School quality, neighborhoods, and housing prices" *American Law and Economics Review* **8** 183–212
- Karsten S, Roeleveld J, Ledoux G, Felix C, Elshof D, 2002, "Schoolkeuze in een multi-etnische samenleving", SCO-Kohnstamm Instituut, Amsterdam

-
- Mulligan C B, 1999, "Galton versus the human capital approach to inheritance" *Journal of Political Economy* **107**(S6), S184–S224
- Murnane R J, Willett R J B, Duhaldeborde Y, Tyler J H, 2000, "How important are the cognitive skills of teenagers in predicting subsequent earnings?" *Journal of Policy Analysis and Management* **19** 547–568
- Propper C, Burgess S, Green K, 2004, "Does competition between hospitals improve the quality of care? Hospital death rates and the NHS internal market" *Journal of Public Economics* **88** 1247–1272
- Staiger D, Stock J H, 1997, "Instrumental variables with weak instruments" *Econometrica* **65** 557–586
- Thomas S, Mortimore P, 1996, "Comparison of value added models for secondary school effectiveness" *Research Papers in Education* **11**(1) 5–53
- Van der Houwen K, Goossen J, Veling I, 2004, "Reisgedrag kinderen basisscholen", Traffic Test bv, Report TT02-95, Commissioned by the Bicycle Council (Fietsberaad), Veenendaal
- Wong H C, Zhan C, Mutter R, 2005, "Do different measures of hospital competition matter in empirical investigations of hospital behavior?" *Review of Industrial Organization* **26** 61–87